

WEEK 2 – POWER BI SALES PERFORMANCE ANALYSIS.

Superstore Sales Performance and Business Insights Dashboard

INTRODUCTION:

Microsoft Power BI is a business intelligence tool that helps users analyze data and present information through interactive reports and dashboards. In this project, the Superstore Sales Dataset was used to study sales, profit, customer orders, product quantity, and regional performance.

The dataset was prepared and organized in Power BI before performing the analysis. A Date Table and relationships were created to support time-based analysis. DAX measures were also developed to calculate important business values.

Different visual elements such as cards, charts, slicers, and a decomposition tree were used to create an interactive dashboard. The completed dashboard makes it easier to understand business performance and identify useful patterns from the sales data.

OBJECTIVES:

The main objectives of this project are:

- *To load the Superstore Sales Dataset into Power BI.
- *To organize the data for analysis.
- *To create a suitable data model.
- *To develop useful DAX measures.
- *To study sales and profit performance.
- *To analyze category and regional performance.

*To examine sales changes over time.

*To create an interactive Power BI dashboard.

*To obtain useful information from the dataset.

DATASET OVERVIEW:

The Superstore Sales Dataset contains details about customers, products, orders, sales, quantity, profit, regions, and customer segments. The dataset can be used to understand how products and customers contribute to overall business performance.

Important Fields;

*Order Date – Date when the customer placed the order.

- *Customer ID – Identifies each customer.
- *Customer Name – Name of the customer.
- *Product ID – Unique product identification.
- *Product Name – Name of the product.
- *Category – Main product category.
- *Sub-Category – Detailed product classification.
- *Sales – Amount generated from sales.
- *Quantity – Number of products purchased.
- *Profit – Profit obtained from sales.
- *Region – Geographical area.
- *Segment – Type of customer segment.

DATA MODELING AND PREPARATION:

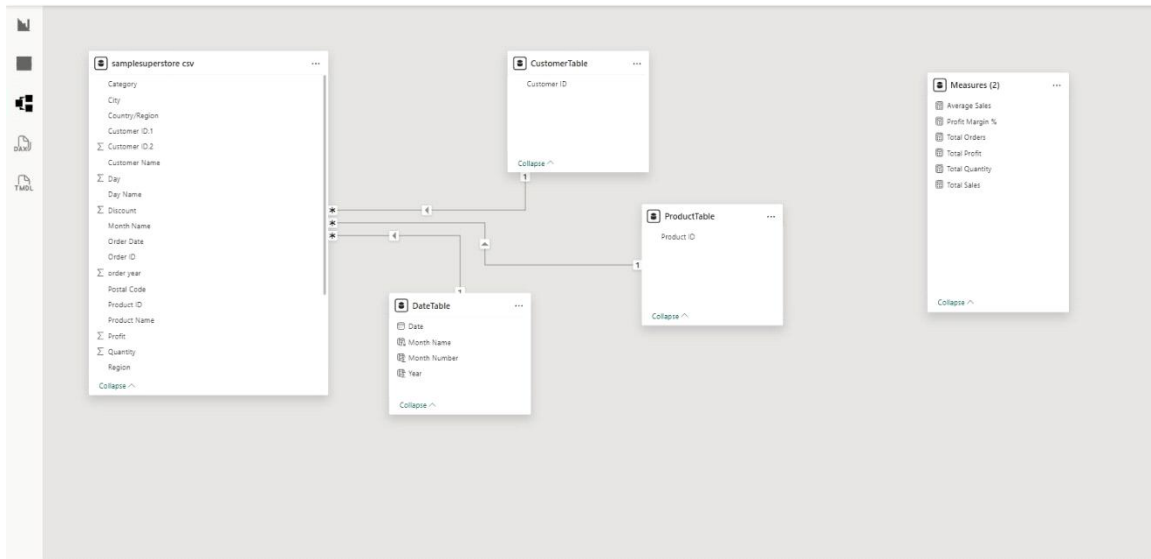
The Superstore dataset was imported into Power BI and examined before

creating the dashboard. The required columns were checked and organized for analysis.

A separate Date Table was created to perform time-based analysis. Additional tables such as Customer, Product, and Region can be used to organize the data model.

Relationships were established between the tables so that information could be analyzed correctly. After preparing the model, DAX measures were created for calculating important business metrics.

Power BI Data Model and Relationships;



TIME-BASED ANALYSIS:

A Date Table was created to analyze sales according to different time periods.

The Date Table contains fields such as:

- *Date.
- *Year.
- *Month Name.
- *Month Number.

The Date Table was connected with the Order Date field of the Superstore dataset. This makes it possible to study sales trends across different years and months.

Date Table in Power BI

The screenshot shows the Power BI interface. In the top-left pane, a DAX formula is defined for a table named 'DateTable':

```
1 DateTable =  
2 CALENDAR(  
3     MIN('samplesuperstore csv'[Order Date]),  
4     MAX('samplesuperstore csv'[Order Date])  
5 )
```

The main view displays the 'DateTable' with the following columns: Date, Year, Month Name, and Month Number. The data represents the first 26 days of January 2023.

Date	Year	Month Name	Month Number
03-01-2023 00:00:00	2023	January	1
04-01-2023 00:00:00	2023	January	1
05-01-2023 00:00:00	2023	January	1
06-01-2023 00:00:00	2023	January	1
07-01-2023 00:00:00	2023	January	1
08-01-2023 00:00:00	2023	January	1
09-01-2023 00:00:00	2023	January	1
10-01-2023 00:00:00	2023	January	1
11-01-2023 00:00:00	2023	January	1
12-01-2023 00:00:00	2023	January	1
13-01-2023 00:00:00	2023	January	1
14-01-2023 00:00:00	2023	January	1
15-01-2023 00:00:00	2023	January	1
16-01-2023 00:00:00	2023	January	1
17-01-2023 00:00:00	2023	January	1
18-01-2023 00:00:00	2023	January	1
19-01-2023 00:00:00	2023	January	1
20-01-2023 00:00:00	2023	January	1
21-01-2023 00:00:00	2023	January	1
22-01-2023 00:00:00	2023	January	1
23-01-2023 00:00:00	2023	January	1
24-01-2023 00:00:00	2023	January	1
25-01-2023 00:00:00	2023	January	1
26-01-2023 00:00:00	2023	January	1

The right-hand pane shows the 'Data' view with a search bar and a list of tables: Measures (2), CustomerTable, DateTable, Date, Month Name, Month Number, Year, ProductTable, and samplesuperstore csv.

DAX CALCULATIONS:

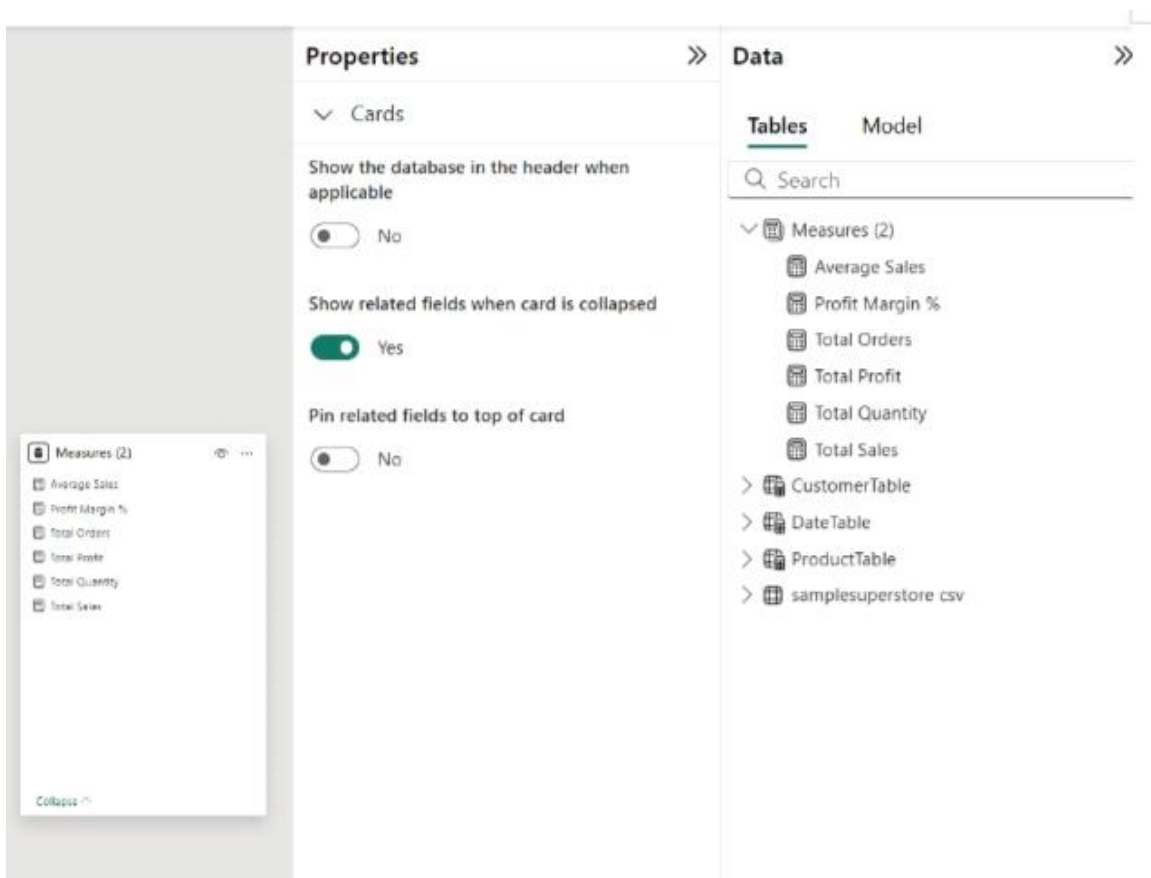
DAX (Data Analysis Expressions) was used to create calculated measures for the dashboard.

The main measures used in the analysis are:

- *Total Sales.
- *Total Profit.
- *Total Orders.
- *Total Quantity.
- *Average Sales.
- *Profit Margin %.

These calculations help convert the raw dataset into useful business information and allow the dashboard to display important values dynamically.

DAX Measures in Power BI.



REPORT VISUALIZATIONS;

Different Power BI visualizations were selected to present the analysis in a simple and understandable format.

1.Sales and Profit KPI Cards.

KPI cards were used to display important overall values such as Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales, and Profit Margin.

These cards provide a quick summary of business performance.

2.Category Sales Comparison.

A column chart was used to compare sales among different product categories. This helps identify differences in sales contribution between categories.

3.Regional Profit Analysis.

A chart was used to compare profit across different regions. It provides a clear view of regional profitability.

4.Monthly/Yearly Sales Trend.

A line chart was created to display sales changes over time. This helps users observe increases or decreases in sales during different periods.

5.Interactive Slicers.

Slicers were added to make the dashboard interactive.

The main filters include:

- *Year
- *Category
- *Region
- *Segment

Users can select different values and view the corresponding changes in the dashboard.

6.Decomposition Tree.

A decomposition tree was included to break down Total Sales based on different categories. It helps users explore the sales data step by step.

DASHBOARD DESIGN:

The final Power BI dashboard combines multiple visualizations into one interactive report.

The dashboard contains:

- *KPI Cards
- *Sales Category Chart
- *Regional Profit Chart
- *Sales Trend Chart
- *Slicers
- *Decomposition Tree

The combination of these components allows users to view important information in one place and apply filters for detailed analysis.

Final Power BI Dashboard



FINDINGS FROM THE ANALYSIS:

The dashboards from the Superstore dataset.

Sales Performance

Total Sales and Average Sales provide an overview of the sales performance.

Profitability

Total Profit and Profit Margin help in understanding the profitability of the business.

Product Category Analysis

Category-wise sales comparison helps identify differences in sales between product categories.

Regional Performance

The regional profit chart helps compare profitability between geographical regions.

Time-Based Performance

The sales trend chart helps observe how sales change across different dates and periods.

Customer Segment Analysis

The Segment slicer allows the dashboard to be viewed according to different customer groups.

Detailed Sales Analysis

The decomposition tree provides a detailed breakdown of Total Sales and allows users to explore the data further.

BENEFITS OF THE DASHBOARD

The Power BI dashboard provides several benefits:

- *Simple Data Understanding** – Complex data can be understood through visual representations.
- *Interactive Reporting** – Users can interact with the dashboard using filters and slicers.
- *Performance Monitoring** – Important sales and profit values can be monitored using KPI cards.
- *Easy Comparison** – Categories and regions can be compared using charts.
- *Trend Identification** – Sales trends can be studied over different time periods.
- *Better Data Analysis** – DAX measures provide useful calculated information.
- *Quick Insights** – Important business information can be viewed quickly from one dashboard.

CONCLUSION

The Superstore Sales Dataset was successfully analyzed using Microsoft Power BI. The data was organized and modeled to support business analysis. A Date Table was created for time-based analysis, and DAX measures were developed to calculate important sales and profit metrics.

Different visualizations, including KPI cards, charts, slicers, and a decomposition tree, were used to develop an interactive dashboard. The dashboard makes it easier to examine sales, profit, category, region, customer segment, and time-based performance.

This project demonstrates how Power BI can be used to convert raw sales data into meaningful visual information that supports business analysis and decision-making.

TOOLS AND TECHNOLOGIES USED

Visualization Tool: Microsoft Power BI.

Data Preparation: Power Query.

Data Modeling: Power BI Relationships and Tables.

Calculation: DAX.

Visualizations: KPI Cards, Column Charts, Line Charts, Slicers, Decomposition Tree.

Dataset: Superstore Sales Dataset.